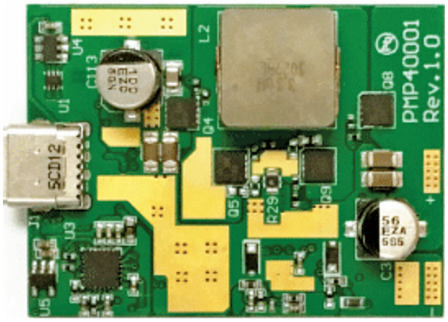
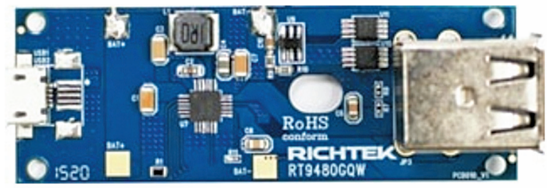
Interesting Reference Designs Of POWER BANKS

Title of Reference Design	USB Type-C Power Bank	Dual Output Power Bank
Image		
Key Attractions	The reference design features three output voltage levels—5V, 12V, and 20V supporting up to 3A each, ensuring compatibility and high performance with up to 96% efficiency for various applications.	The power management solution simplifies portable device charging by integrating a linear charger, dual-output management, torch function, and efficient power boost capabilities to enhance user experience and device safety.
Highlights	The PMP40001 is a reference design by Texas Instruments for a USB Type-C power delivery downstream facing port, crafted to meet contemporary power management needs. The design offers three output voltage levels—5V, 12V, and 20V with each supporting a maximum output current of 3A. This capability allows the design to supply up to 60W of power at 20V, catering to various applications with different power requirements. Achieving up to 96% efficiency at full load, this high performance is due to the LM5175 buck-boost controller, which enhances power efficiency and thermal performance—key for the reliability and longevity of devices. Operating with a wide input voltage range of 6V to 13.5V, the design is compatible with 2S and 3S lithium battery packs. Its compact size and low component count make it suitable for space-constrained applications without sacrificing performance. The design supports high discharge power up to 20V/3A, aligning with USB Type-C PD profile 4 for broad device compatibility.	The RT9480 from Richtek is a comprehensive, integrated power management solution designed specifically for Li-ion power banks and other portable handheld devices. This user-friendly product, known as EZPBS (easy to use power bank solution), includes a linear charger, a synchronous boost converter with dual output load management, and a torch function, all in a single chip. It features a 4-LED indicator for displaying battery capacity and the charging or discharging status. The design ensures safety and efficiency with protection functions like over-temperature protection (OTP), over-voltage protection (OVP), over-current protection (OCP), and output short protection. It supports up to 1.2A charging with features like dynamic power management (DPM), thermal regulation, auto-recharge, and JEITA function support. The device's USB output can handle dual USB outputs with auto and button control, and its synchronous boost can deliver up to 2.5A with a peak efficiency of 97%.
Applications	The reference design is suitable for portable electronics and industrial applications.	The reference design is designed for use in Li-ion power bank applications.
OEM Brand	Texas Instruments (TI)	Richtek
URL	https://www.electronicsforu.com/electronics-projects/usb-type-c-power-bank-reference-design-2	https://www.electronicsforu.com/electronics-projects/dual-output-power-bank-reference-design